

CO4_AT1 – DATA INTERPRETATION

Expanded 10-Mark Answers

1. Real-Time Detection System

A real-time surveillance system must detect objects such as vehicles, pedestrians, or suspicious activities quickly. Therefore, accuracy alone cannot determine the best model. The system must consider both detection accuracy and processing speed.

Method	Accuracy	Speed
Viola-Jones	78%	Fast
YOLO	92%	Moderate
Deep Learning	95%	Slow

Viola-Jones has an accuracy of 78% and provides fast processing. Its main advantage is low computational complexity, making it suitable for systems with limited hardware resources. However, its lower accuracy can lead to missed detections or false detections, particularly in complex scenes.

YOLO achieves 92% accuracy with moderate processing speed. It provides a strong compromise between detection accuracy and computational efficiency. Its accuracy is 14 percentage points higher than Viola-Jones while remaining faster than the slow Deep Learning option in the given comparison.

Deep Learning provides the highest accuracy at 95%. However, its slow processing speed can create unacceptable latency when decisions must be made immediately. It may also require more powerful GPUs or hardware acceleration.

Parameter	Viola-Jones	YOLO	Deep Learning
Accuracy	78%	92%	95%
Speed	Fast	Moderate	Slow
Computational requirement	Low	Moderate	High
Detection quality	Moderate	High	Very high
Real-time suitability	High	Very high	Moderate
Complex scenes	Limited	Good	Very good

If the system has extremely strict latency constraints, Viola-Jones can be selected because it is the fastest. However, its 78% accuracy may not be acceptable for reliable surveillance. If moderate latency is allowed, YOLO is the most practical choice because it gives much higher accuracy while remaining faster than Deep Learning.

Deployment conditions: Low-power device and strict latency → Viola-Jones. Normal real-time surveillance → YOLO. High-performance hardware with accuracy priority → Deep Learning.

Final evaluation: YOLO is the recommended overall method for a real-time surveillance system because it provides the best practical balance between 92% accuracy and moderate speed. Viola-Jones is preferred only when latency is the overriding constraint, while Deep Learning is preferred when maximum accuracy is more important than response time.

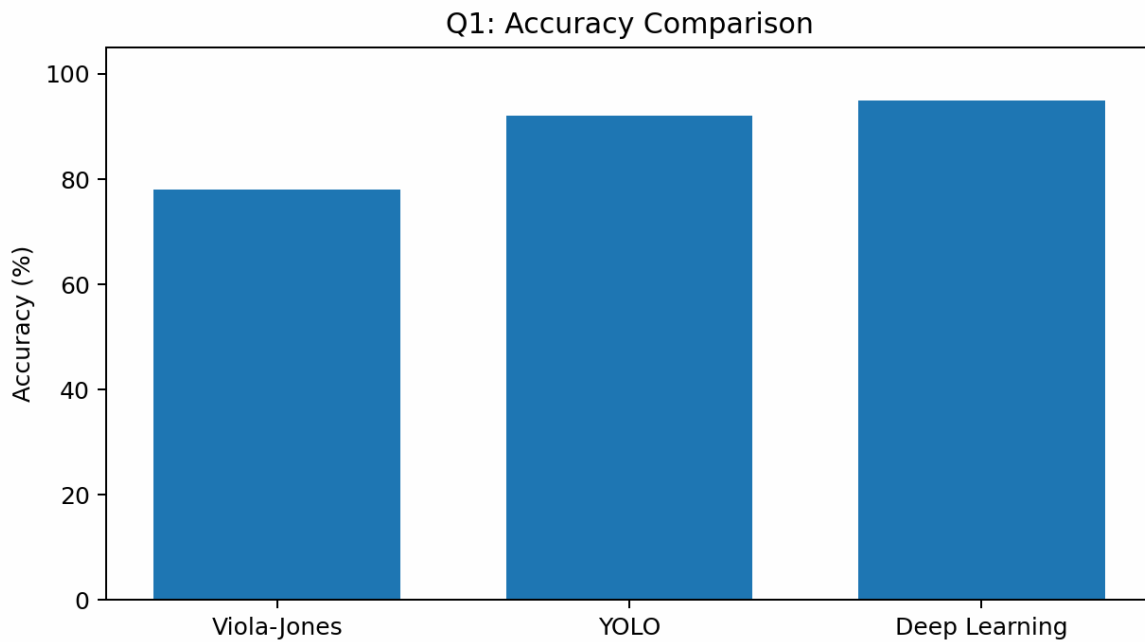


Figure 1: Accuracy comparison of the three detection methods.

2. Robust Object Detection

A robust object detection system should continue to perform accurately when environmental conditions change. The experiment evaluates normal conditions, occlusion, and low-light conditions.

Method	Normal	Occlusion	Low Light
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

Under normal conditions, Deep Learning achieves 95%, YOLO 90%, and Viola-Jones 80%. Under occlusion, all methods lose accuracy because part of the object is hidden. Viola-Jones falls to 60%, YOLO to 75%, while Deep Learning remains at 85%. In low light, Viola-Jones reaches only 55%, YOLO 70%, and Deep Learning 80%.

Method	Normal	Occlusion	Drop	Low Light	Drop
Viola-Jones	80%	60%	20 points	55%	25 points
YOLO	90%	75%	15 points	70%	20 points
Deep Learning	95%	85%	10 points	80%	15 points

Average accuracy: Viola-Jones = $(80+60+55)/3 = 65\%$. YOLO = $(90+75+70)/3 = 78.33\%$. Deep Learning = $(95+85+80)/3 = 86.67\%$. Therefore, Deep Learning has the highest average performance and the smallest degradation.

Viola-Jones is fast and computationally efficient, but it is less robust. YOLO provides a good balance between accuracy and speed. Deep Learning provides the strongest performance across difficult conditions but generally requires more computation and may increase latency.

Final evaluation: Deep Learning provides the best overall robustness, with an average accuracy of approximately 86.67%. It should be selected when reliable detection under occlusion and low light is the main requirement. YOLO is a strong alternative when real-time speed and resource limits are important.

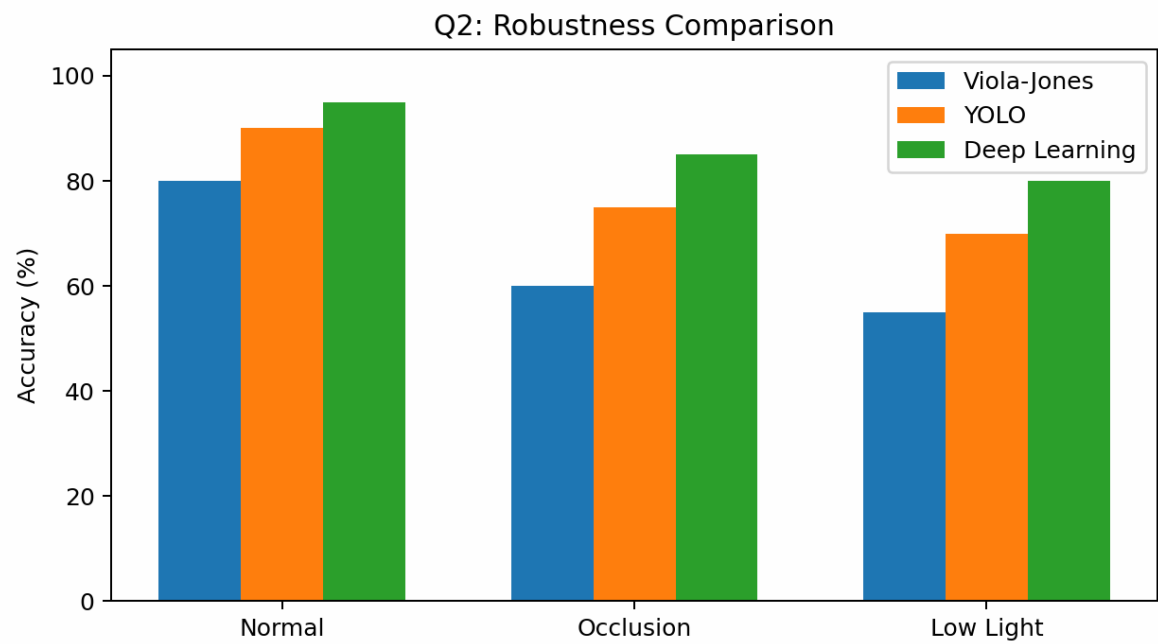


Figure 2: Robustness comparison under normal, occluded, and low-light conditions.

3. Optical Flow Analysis

Optical flow estimates the apparent movement of pixels or objects between consecutive video frames. It is used in object tracking, vehicle tracking, pedestrian tracking, surveillance, gesture recognition, and motion analysis.

Method A produces stable motion vectors but misses small motions. Its major strengths are stability, low noise sensitivity, smooth trajectories, and reliable long-term tracking. Its limitation is reduced sensitivity to subtle movements.

Method B detects fine motion but produces noisy vectors. It is useful for small or subtle movement analysis, but noisy vectors can create false motion and unstable tracking. Additional smoothing, filtering, or temporal consistency checks may be required.

Criterion	Method A	Method B
Motion stability	High	Low
Small-motion detection	Low	High
Noise sensitivity	Low	High
Tracking reliability	High	Moderate
Fine motion analysis	Limited	Excellent
Post-processing requirement	Low	High

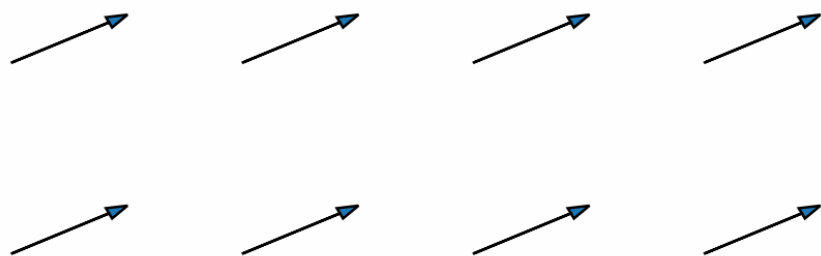
For general dynamic-scene motion tracking, Method A is preferable because stable motion vectors are important for maintaining object identity across frames. For example, traffic surveillance benefits from smooth and reliable trajectories.

Method B may be more appropriate for applications such as hand movement detection, facial movement analysis, small object motion, medical motion analysis, and fine industrial motion. A practical system can combine Method B with filtering and temporal smoothing to improve stability.

Final evaluation: Method A is more suitable for general-purpose dynamic-scene tracking because it provides stable and less noisy motion vectors. Method B should be selected when detecting small or subtle movements is the main objective.

Q3: Optical Flow Comparison

Method A: stable motion vectors



Method B: fine motion but noisy vectors

Figure 3: Conceptual comparison of stable and noisy optical-flow vectors.

4. Model Generalization and Overfitting

Model A achieves 98% training accuracy but only 70% testing accuracy. The difference is 28 percentage points. This large gap indicates that the model has likely learned specific patterns or noise from the training data instead of learning general patterns. This is a strong indication of overfitting.

Model B achieves 90% training accuracy and 88% testing accuracy. Its difference is only 2 percentage points, indicating much better generalization and a lower risk of overfitting.

Parameter	Model A	Model B
Training accuracy	98%	90%
Testing accuracy	70%	88%
Accuracy gap	28 percentage points	2 percentage points
Overfitting risk	High	Low
Generalization	Poor	Good
Real-world reliability	Low	High

In real-world applications, the model encounters new data that was not present during training. Therefore, testing performance is more important than training performance. Model B has 88% testing accuracy, which is considerably higher than Model A's 70%.

Model A could potentially be improved using more training data, data augmentation, regularization, dropout, cross-validation, early stopping, feature selection, or reduced model complexity.

Final evaluation: Model B is more reliable for real-world deployment. Its testing accuracy is high and close to its training accuracy, showing consistent performance on unseen data. Model A should not be selected without further measures to control overfitting.

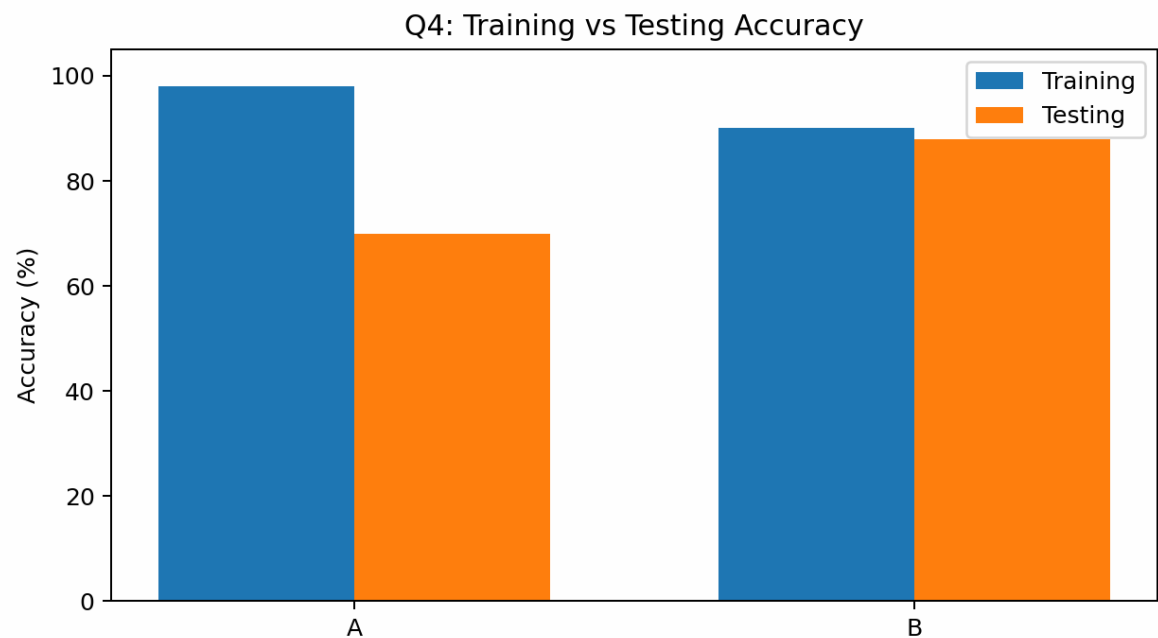


Figure 4: Training and testing accuracy showing the overfitting risk of Model A.

5. System Performance under Constraints

A real-time application must consider accuracy, latency, and cost simultaneously. A highly accurate model may not be useful if its latency is too high for the application.

Model	Accuracy	Latency	Cost
A	88%	50 ms	Low
B	92%	80 ms	Moderate
C	95%	120 ms	High

Model A provides 88% accuracy, 50 ms latency, and low cost. Its main advantage is speed and affordability. It is appropriate when fast response and low deployment cost are important, but its accuracy is lower than the other models.

Model B provides 92% accuracy, 80 ms latency, and moderate cost. It improves accuracy by 4 percentage points over Model A while adding 30 ms of latency. This represents a practical compromise for many real-time applications.

Model C provides 95% accuracy, 120 ms latency, and high cost. It has the highest accuracy, but compared with Model A it adds 70 ms of latency and significantly increases deployment cost.

Parameter	Model A	Model B	Model C
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Accuracy	88%	92%	95%
Latency	50 ms	80 ms	120 ms
Cost	Low	Moderate	High
Speed	Highest	Good	Lowest
Accuracy level	Moderate	High	Highest
Overall balance	Good	Best	Expensive

Choose Model A when very low latency, limited hardware, and low cost are dominant. Choose Model B when good accuracy, moderate latency, and reasonable cost are required. Choose Model C when maximum accuracy is critical and higher latency and cost can be accepted.

Final evaluation: Model B provides the best balance between accuracy, latency, and cost. It offers 92% accuracy with 80 ms latency and moderate cost, making it the recommended choice for a typical real-time computer-vision application.

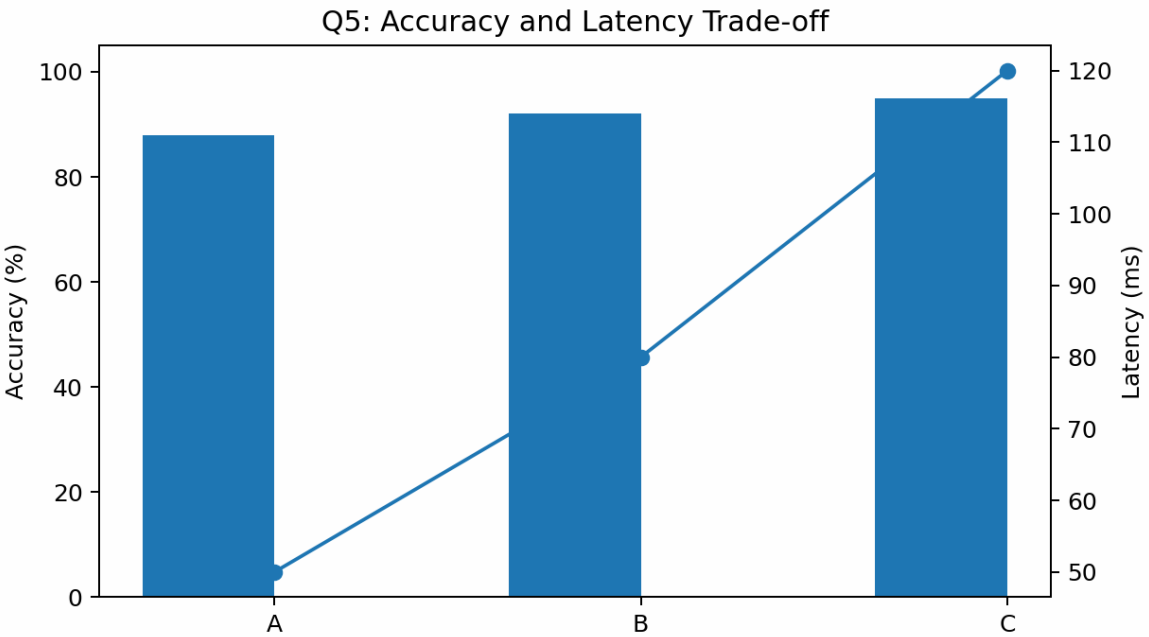


Figure 5: Accuracy and latency trade-off among the three models.

Overall Conclusion

The analysis demonstrates that computer-vision model selection should not depend on accuracy alone. Real-world deployment requires a balance among accuracy, robustness, latency, computational cost, and generalization. YOLO is a strong compromise for real-time detection, Deep Learning provides the highest robustness, Model B in Question 4 has the best generalization, and Model B in Question 5 provides the best overall accuracy-latency-cost balance.